

# Coverage Report

**Measurement:** Foundry line, statement, branch, and function coverage

**Scope:** Production contracts in `src/`, excluding mocks, tests, and scripts

**Requirement:** At least 90% line coverage across contracts

**Status:** Passing with 92.31% line coverage

## 1. Command

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```
forge coverage --ir-minimum --report summary --no-match-coverage 'script|test|src/mocks'
```

`--ir-minimum` is used because Foundry coverage disables optimizer/viaIR by default, and the large deployment script otherwise hits Solidity stack-depth limits. The same command is documented for local review and CI consistency.

## 2. Coverage Summary

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| Scope                                                     | Lines            | Statements       | Branches       | Functions      |
|-----------------------------------------------------------|------------------|------------------|----------------|----------------|
| Production contracts in <code>src/</code> excluding mocks | 92.31% (276/299) | 91.24% (250/274) | 34.41% (32/93) | 94.81% (73/77) |

### 3. Test Suite Evidence

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**88**

Foundry tests currently passing.

**50+**

Unit tests cover public/external functions and revert paths.

**10+**

Fuzz tests cover AMM, ERC4626, and governance paths.

Additional coverage classes:

- Invariant tests cover AMM constant product behavior, ERC20 supply conservation, vault accounting, treasury accounting, and governance role safety.
- Fork hooks cover USDC, Uniswap V2 router, and Chainlink feed integrations when `MAINNET_RPC_URL` is present.
- Vulnerability case studies reproduce and fix reentrancy and missing access control.

## 4. Notes for Final Submission

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Scripts, tests, and mock contracts are excluded from the graded production-contract denominator. If final code changes add new production branches, rerun coverage and update this report before submission.